Hands-On with the Adafruit CircuitPlayground Bluefruit

High school visit (04/20/2026)

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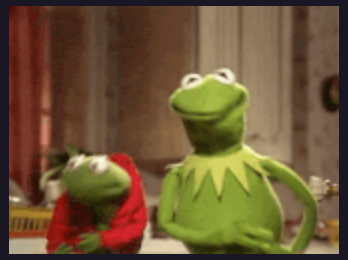
https://efredericks.github.io



Plan for today

1. **Block programming in a simulator**
2. **Program device**
3. **Try it out on the speakers!**

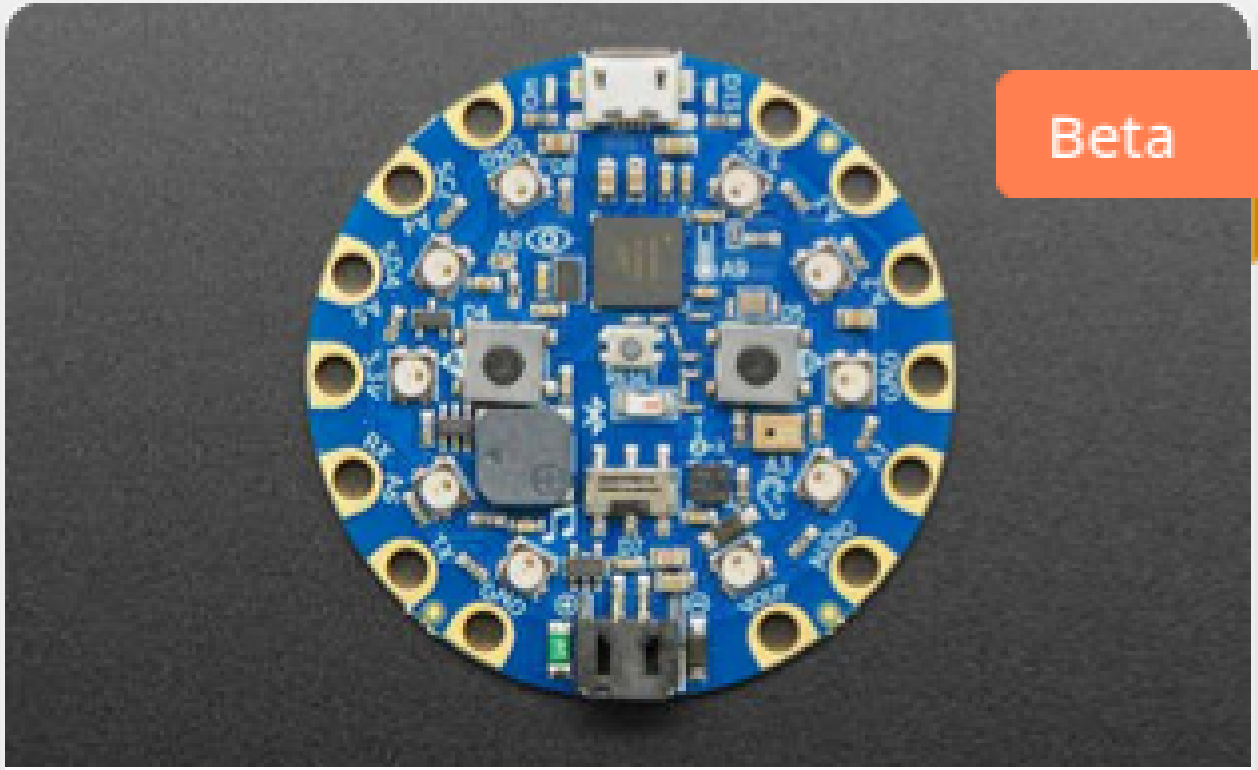
Block programming

In a browser, go to:

https://maker.makecode.com

First steps

1. **Click 'New Project'**
2. **Give it a name!**
3. **Click create and then select the adafruit-circuit-playground-bluefruit**



adafruit-circuit-playgro...

The Adafruit Circuit Playground Bluefruit library - beta

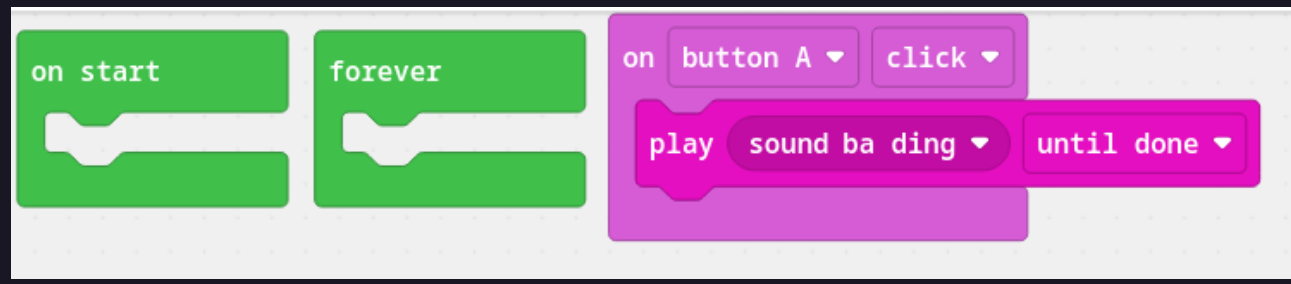
Make a sound!

Let's make a sound on a button press.

Click `Input` and get an `on button A click` field.

- Anything in that block will run when you click a button
- You can put any action in there!

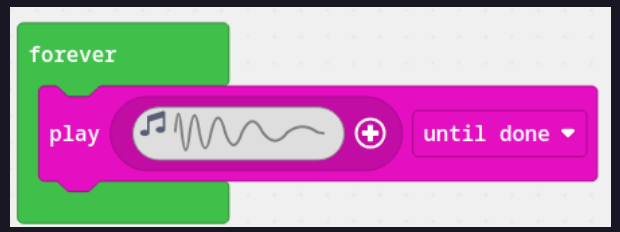
Now, pop a sound in. Click `Music` and then `play sound ba ding until done` and drag that *inside* the button event.

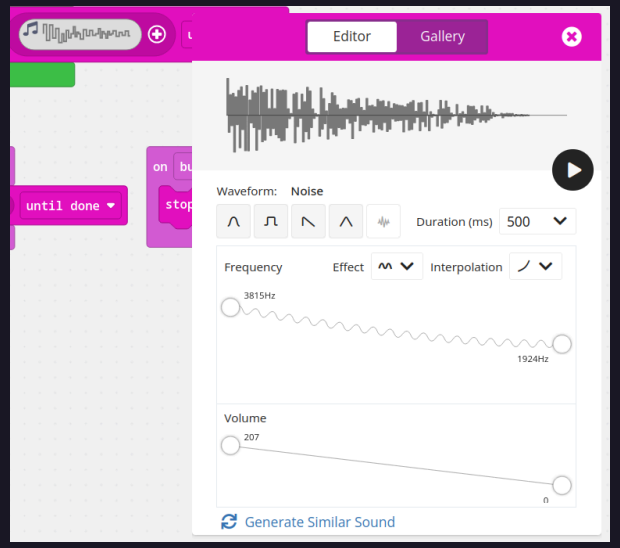


Download to the device!

This is all on your computer - let's put it on a device!

1. **Click the `Download` button - a `*.uf2` file will download to your computer.**
2. **Make sure the device is plugged in! If you don't see a `CPLAYBTBOOT` folder in your file explorer then click the small button in the middle of to reset it.**
   - If it still doesn't work, try clicking it twice to put it into bootloader mode.
3. Drag the `.uf2` file onto your `CPLAYBTBOOT` directory
   - It should auto-load - if not try resetting it!





Now some forever sounds!

Add the `play <waveform>` node to your `forever` block

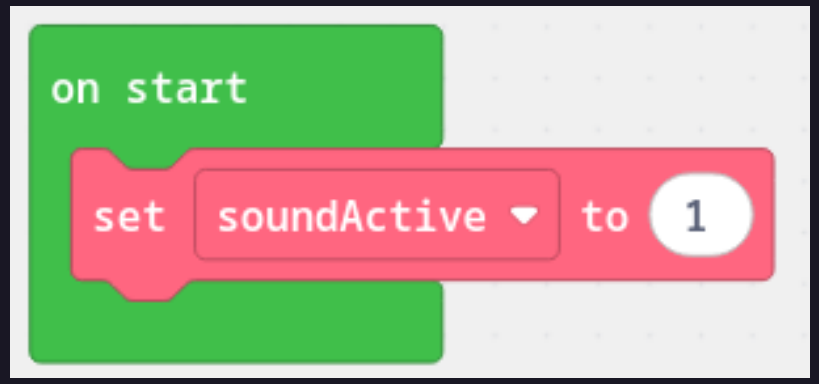
- Play with the waveform by clicking on it and moving the points around!

Now make a variable and turn it on/off!

First add a variable - click `Variables` **and** `Make a variable` **and call it** `soundActive`

Then, drag the `set soundActive to 0` **field to the** `on start` **block**

- Set its value to be `1`

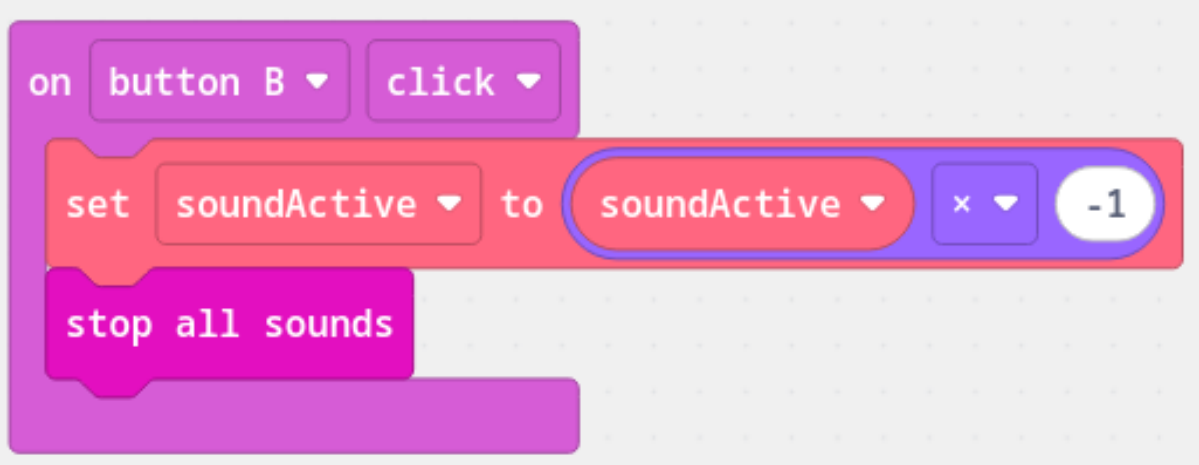


And change the variable

Add another button block and have it respond to Button B.

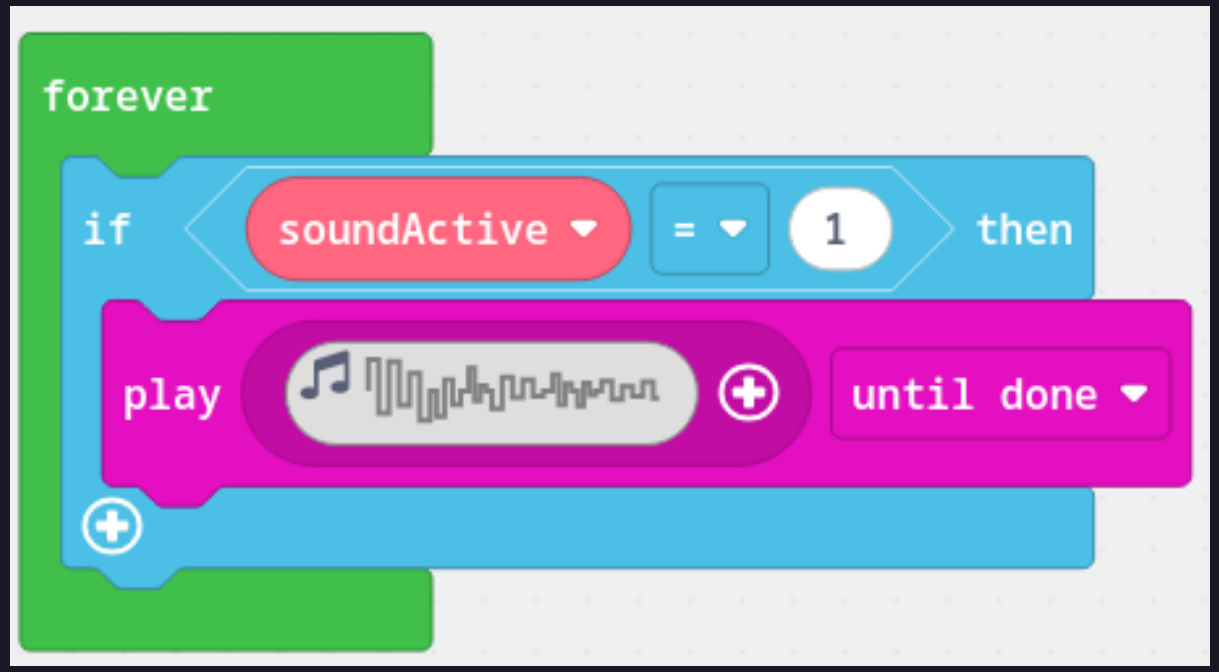
Inside that block, change the `soundActive` variable to be the opposite!

Also, add the `stop all sounds` block so that things stop!



And then, the if statement

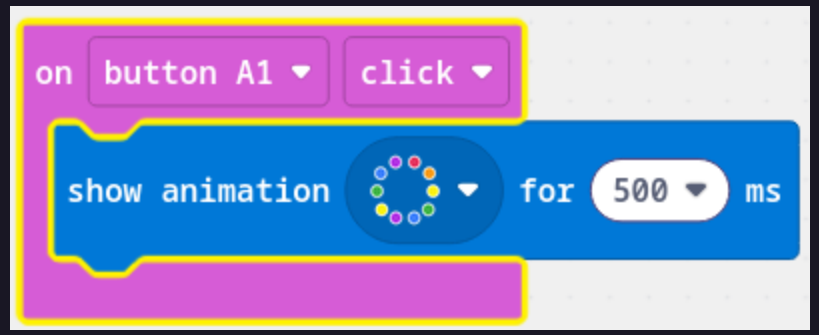
In your `forever` block, add an `if` statement to play sound! This will involve adding a bunch of things, so good luck!



Create a lightshow on touch

Add another input, this time for a touch sensor

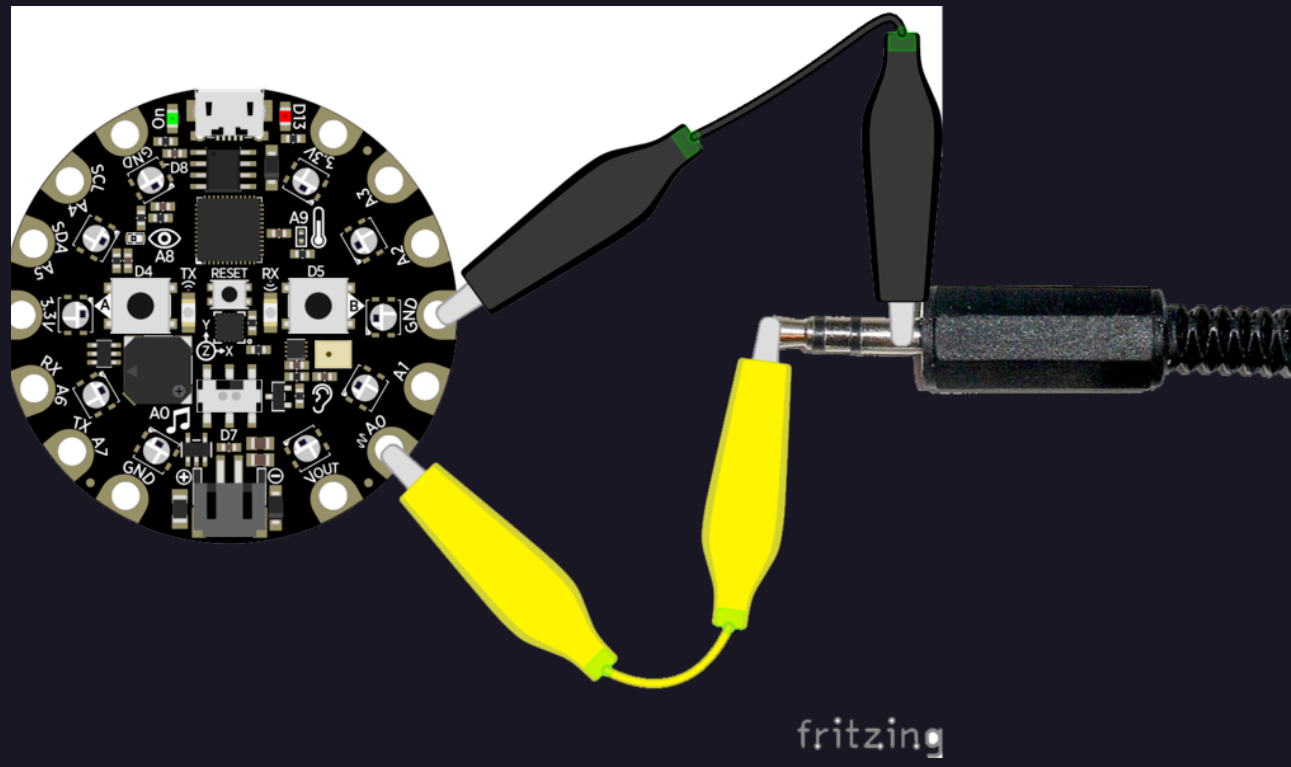
Then, add a lightshow!

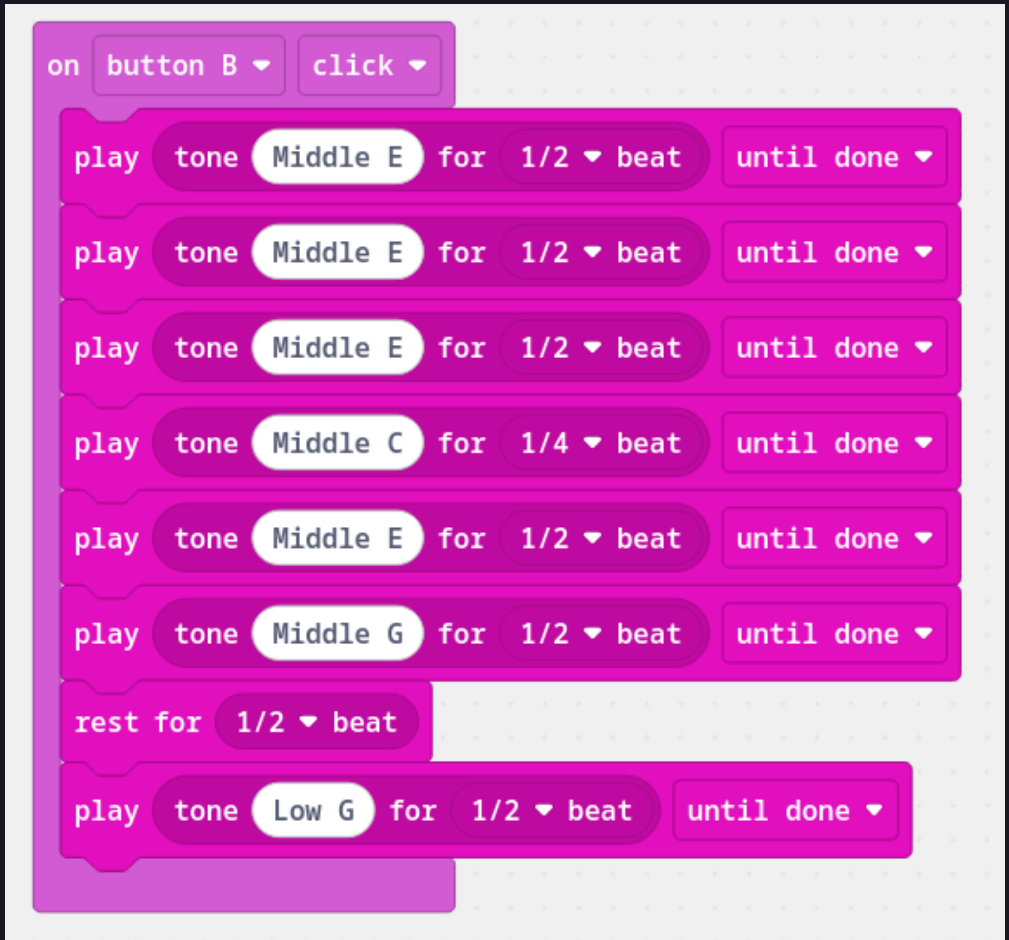


Hook up to a speaker!

If you want to test out your program on a *better* speaker, here's how it can be wired up!

- Connect the `GND` pad (ground) to the base of the audio connector
- Connect the tip of the audio connector to the `AUDIO` pad





Have some fun!